Biomechanics of the Basketball Jump Shot—

Six Key Teaching Points

Based on the biomechanics research on shooting a basketball, six key teaching points for mid-range to long-range jump shots are proposed. These concepts should be evaluated and translated into appropriate cue words for feedback to performers.

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Teachers and coaches interested in basketball should integrate ideas from practical experience and biomechanics research so that their teaching elicits the best performance from their athletes. A focus of biomechanical research has been to identify the critical elements in the jump shot that are related to shot success. A previous review of the jump shot (Satern, 1988) suggested that the shooting styles of the free throw, set, and jump shots should be similar. This article reviews the biomechanics research on the basketball jump shot and free throw to identify key technique points of the jump shot for instructors to use to improve player performance.

These critical elements will be discussed in relation to the shooter's body position, ball release velocity, and ball spin.

Body Position

For a given shooting position, an athlete will release the ball with a velocity appropriate to move the ball horizontally and vertically to the goal. Coaching literature is fairly consistent in advising athletes to "square up" the body to the basket and jump vertically. A stable base of support would minimize horizontal motion of the body and create an accurate shot. The research suggests that coaches focus on a *staggered stance and a vertical jump* in teaching the jump shot.

Ryan and Holt (1989) observed a statistical relationship between a staggered stance with the feet aligned between 14 and 18 degrees from parallel and jump shooting success. They suggested that squaring up the feet in a parallel stance is not critical. Ingram and Snowden (1989) suggested that "face up" might be a better teaching cue for shooting because skilled subjects stagger their stance and align their trunk to position their shooting shoulder forward. Based on a review of research, Martin (1981) recommended emphasizing a slightly staggered and narrow (10 inches) stance, and a jump from a moderate (down to 115-120 degrees of knee flexion) countermovement. A base of support slightly less than shoulder width, slightly staggered with the shooting side foot forward, and with both feet slightly pointed outward provides the most stable base for the jump shot. The slightly staggered stance allows the performer to minimize forward or backward motion of the body in the jump, and maintain side-to-side alignment of the body with the basket.

Highly skilled shooters have been observed to have less horizontal shift in their center of gravity during the shot than less skilled shooters (Drysdale, 1973; Hudson, 1982, 1985a, 1985b; Satern, Messier, & Keller-McNulty, 1989; Yates & Holt, 1982). Some studies have reported small forward movements in skilled adult players for long jump shots (Gaunt, 1976; Penrose & Blanksby, 1976; White & Elliott, 1989). Some forward motion in the jump shot may be necessary for long shots. Because this motion is very small in normal range shooting, and accuracy would be improved with a stable base, athletes should minimize horizontal motion by jumping as close to vertical as possible.

Several authors suggest that shot accuracy will be best if the shooter keeps the ball, wrist, elbow, and shoulder in a plane aligned with the basket (Hartley & Fulton, 1971;

Hudson 1985c; Ingram & Snowden, 1989; Szymanski, 1967). Maintaining the shooting side of the body in this *shooting plane* aligned with the basket would be a desirable result of minimizing horizontal motion in the jump. Ingram and Snowden (1989) have suggested that the traditional square up stance makes it difficult to use correct shooting form and align the shooting forearm with the basket. Shot accuracy will be enhanced by a staggered stance that allows the athlete to align the shooting arm with basket, and easily sight along this *shooting plane.*

Several scholars have convincingly shown that maximizing the height of release increases shot accuracy and increases the margin of error for the shot (Brancazio, 1981; Hay, 1985; Mortimer, 1951). An *optimized height of release* decreases the distance the ball must travel, decreases the ball velocity needed to execute the shot, and will provide a more favorable angle of entry. In the free throw, increased plantar flexion that would increase the height of release has been statistically related to shot success (Ryan & Holt, 1989). Teachers should examine three technique points that contribute to maximizing the height of release: (1) a balanced vigorous jump; (2) the timing of release; and (3) shoulder flexion prior to release.

The uniqueness of the jump shot is the controlled jump that lifts the performer prior to releasing the ball. The jump needs to be vigorous, yet balanced. Elliott and White (1989) measured peak vertical ground reaction forces of 2.7 times body weight (BW), while peak horizontal ground reaction forces of 0.5 BW were used to minimize forward motion in the jump shot. It is clear that the controlled, vigorous jump in skilled jump shooting generates large forces similar to the forces in running.

Instructors should focus on the inclination of the trunk during the upward motion of the jump shot because the trunk has 50 percent of the body mass and will dramatically affect balance. Research on shooting a basket has demonstrated that skilled shooters keep the trunk within a few degrees of erect near release (Hudson, 1985a, 1985b; Penrose & Blanksby, 1976; Ryan & Holt, 1989; Satern et al. 1989; Szymanski, 1967; White & Elliott, 1989). Once the athlete has left the ground, the trunk should remain in a nearly vertical position; excess forward or backward trunk lean should be eliminated as this will change the appropriate angle of release (Elliott & White, 1989; Satern et al. 1989) and decrease the height of release. "Fade away" jump shots are undesirable because of the inconsistent position and velocity of the body at release, the unstable position of the body, and the decreased height of release.

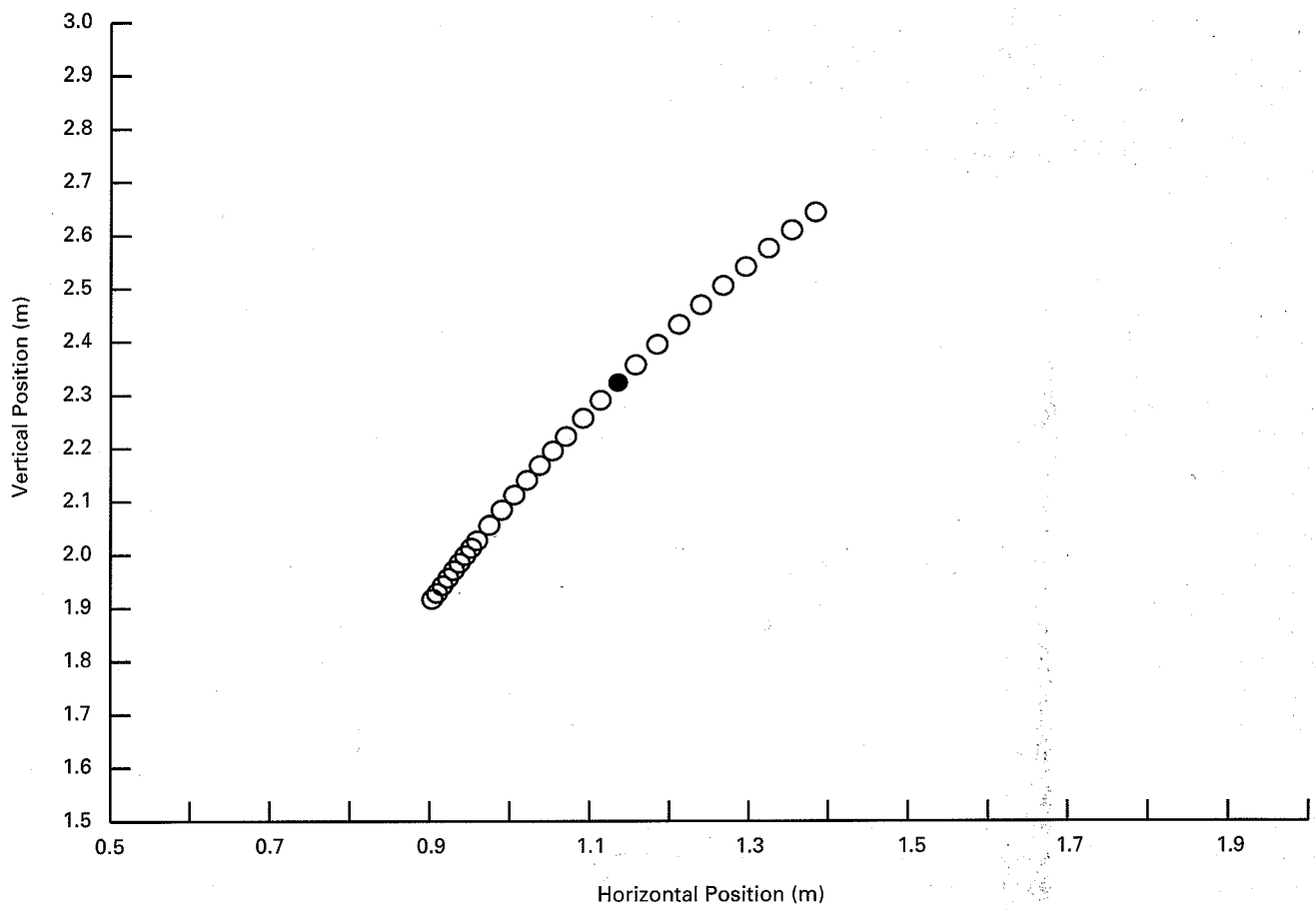

Figure 1. Trajectory of a 15-foot shot for the desirable angles of release, and the range of physically possible angles. (From Hay, 1985, p. 222. Copyright 1985 by Prentice-Hall, Inc. Reprinted by permission.)

The timing of release near the peak of the jump would also tend to increase the height of release. Several studies have documented that release occurs just prior to the peak of the jump (Drysdale, 1973; Elliott & White, 1989; Martin, 1981; Szymanski, 1967; Walters, Hudson, & Bird, 1990). It is not clear if the vertical velocity of the body is a significant contributing factor to ball velocity. Some studies suggest that vertical velocity of the body contributes to the ball's vertical velocity (Tsarouchas, Kalamaros, Giarroglou, & Prassas, 1990; Walters et al. 1990). Another study noted that timing release near the peak of the jump is advantageous because the vertical velocity of the body is near zero (Elliott & White, 1989).

Research has shown that the range of shoulder flexion is related to shot success (Yates & Holt, 1982), which may be related to the increased height of release. The angle of the humerus at release in a free throw is approximately 40 degrees above the horizontal (Satern et al. 1989;). White and Elliott (1989) reported that shoulder flexion at release increased (15 degrees) as female jump shooters moved from 13 to 20 feet from the basket. Athletes should be encouraged to increase shoulder flexion of the shooting arm, possibly by using less than anatomically accurate cues such as "extend the arm upward."

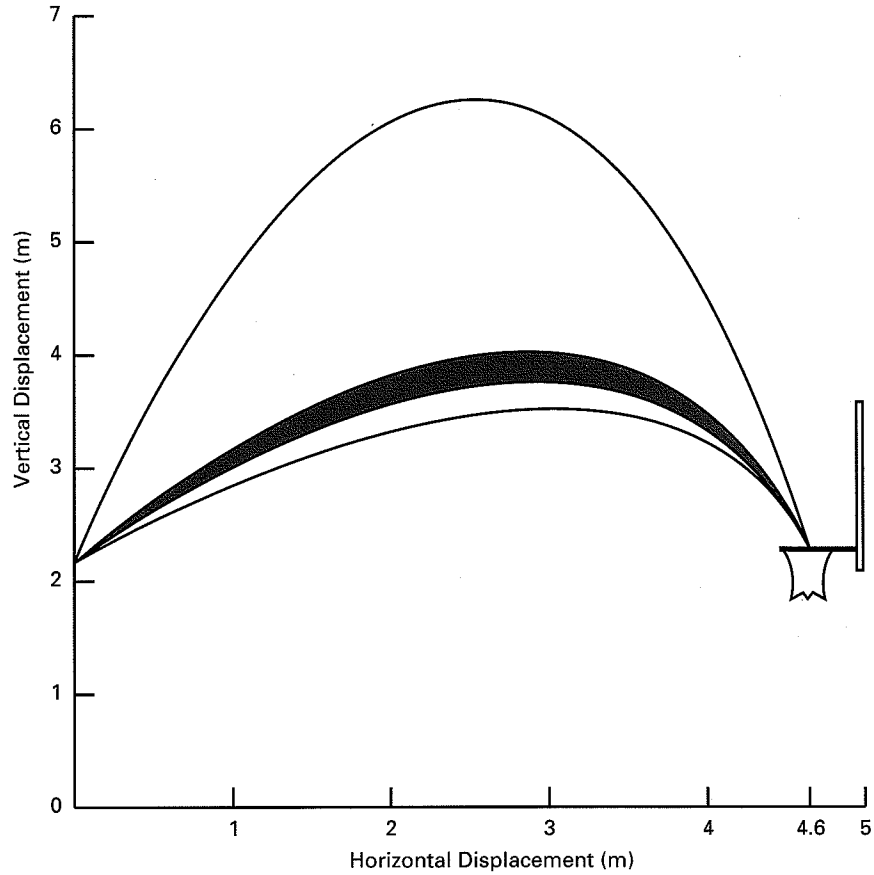

Figure 2. Motion of the basketball near release (side view) during a free throw by a male collegiate player (career 78%).

Release Velocity

The velocity of the ball at release may be the most significant factor determining shot success. Theoretical studies have shown that for an appropriate angle of release, error tolerance in the correct ball speed is about 1 percent (Brancazio, 1981; Mortimer, 1951). One biomechanical study examined successful and unsuccessful shots from a variety of distances and found that 65 to 85 percent of the errors were a result of initial ball velocity (Toyoshima, Hoshikawa, & Ikegami, 1981). Since velocity is speed in a specific direction, ball speed and release angle combine to determine the velocity of ball release. These two factors interact because a minor variation in one will require a change in the other for the shot to be successful. There are, in fact, a variety of appropriate release angles and ball speeds for a given shot (Brancazio, 1981; Elliott & White, 1989; Hay, 1985; Walters et al. 1990).

Biomechanical research has reported the ball speeds at release for many types of basketball shots. Ball speeds at release in free throws have been reported between 6.0 and 7.2 m/s (Hudson, 1982, 1985a; Satern, 1990; Satern et al. 1989), while jump shots from approximately the same distance have ball speeds between 6.6 and 6.9 m/s (Walters et al. 1990).

The data on ball speeds at release are consistent with Brancazio's theory (1981), that good shooters select the release angles that are very near the angle that allows a minimum speed for a successful shot. Shooting at the minimum speed has several advantages: increased accuracy, limiting fatigue, and maximizing the possibility that the ball will pass through the basket.

Since ball speed is difficult for teachers to evaluate and will interact with the release angle, coaches should focus on the *angle of release* and the technique or coordination of joint actions to improve shooting skill. The release angle of a jump shot has a dramatic effect on the shot's success. Some release angles will not permit an angle of entry that allows the ball to pass cleanly through the basket (Hay, 1985). There is not one perfect angle to release a jump shot because a player's position on the court and height of release vary. Research has shown that factors such as court position, height of release, shoulder angle, and trunk angle affect the angle of release (Brancazio, 1981; Elliott & White, 1989; Satern et al. 1989). The research, however, has indicated that there is a desirable range of release angles that are re-

Table 1. Angles of Release for Basketball Shots Between 10 and 21 Feet

Source	Angle(s)	Evidence
Brancazio, 1981	45-55	Theory
Hay, 1985	49-55	Theory
Mortimer, 1951	54-58	Theory
Hudson, 1985a	53	Female free throws
Tsarouchas et al., 1988	50	Male free throws
Satern, 1988	52-55	Free throws
Hudson, 1982	52	Female free throws
Walters et al., 1990	52	Female jump shots
White & Elliott, 1989	53	Female jump shots
Toyoshima et al., 1981	51-55	Male jump shots

lated to shot success. Table 1 lists the angles of release theoretically calculated or measured for midrange (10-21 feet) basketball shots.

Table 1 suggests that the experimental data agree with the theoretical predictions of optimal release angles. Hay (1985) proposed midrange jump shots be projected between 49 and 55 degrees. Based on the experimental data and theory, an *angle of release* of 52 degrees would be an appropriate guideline. Shooting at angles near 52 degrees provides shots with near-minimum ball speeds and desirable angles of entry. Balls shot with these trajectories have a better chance of passing through the basket than shots with a flatter release angle. Shots closer to the basket or shots from shorter performers will require release angles in the upper range of desirable angles (Brancazio, 1981; Hudson, 1985c).

Coaches and teachers should observe some trials in shooting practice from a side view, at least 10 meters from the performer. Shots should be taken with initial trajectories just above an imaginary 45-degree angle, and would then approximate the 49- to 55-degree optimal range. Sagittal plane video images could allow coaches to make crude measurements from the screen to check if shooters are using appropriate angles. Some authors have advocated that trajectory feedback to performers be made regarding the peak of flight of the shot (Hofer, 1978; Mortimer, 1951; Mullaney, 1957) to achieve the appropriate release angles for free throws. For a free throw, the high point of the slightly nonsymmetrical trajectory (figure 2) is 12.4 feet (Tan & Miller, 1981) and occurs two-thirds of the way to the basket (Hofer, 1978). This could be a valuable visual checkpoint for instructors and athletes.

A common cue for teaching basketball shooting is to use a "high arch" for the shot. Experienced coaches know that novice performers use lower than appropriate release angles (Hudson, 1985c), shooting the ball at the front of the rim rather than upwards above the head. Figure 1 illustrates the path of the ball for a 15-foot shot. Hay (1985) used this figure to illustrate how appropriate release angles are not really "high." Few shots in actual play are projected above 60 degrees (Brancazio, 1981). This review suggests that instructors use whatever cues (high, up, moderate) that help performers obtain the correct angle of release.

The shooting arm technique that creates ball velocity in the jump shot seems to be clearly defined in teaching and coaching literature (Cooper & Siedentop, 1969; Cousey & Power, 1970; Hess, 1980; Wooden, 1988). Unfortunately, research on shooting the jump shot has not clearly shown what pattern of coordination skilled players use, or what pattern is most desirable. Currently, the most critical issue is the integration or *cooperation of the upper and lower extremities.*

Hudson (1985c) advocated that the coordination in free throw shooting be a smooth integration of motions of the legs, arm, forearm, and hand. She recommended using a shooting technique that combined moderate velocities from a minimal number of segments, which would be relative to the strength of the performer. The hypothesis that strength is related to joint coordination or the range of motion employed (Hudson, 1985c; Satern, 1990) has been researched with equivocal results (Elliott & White, 1989; Pease, 1981). Satern (1990) found that skilled adult males and females used more compact free throw motions compared to adolescents who used more simultaneous actions of the joints of the body. More research on this aspect of shooting coordination is needed because the strength of performers varies. Some studies have examined the motion adjustments of children using smaller basketballs and lower baskets (Juhasz & Wilson, 1982; Satern et al. 1989).

Biomechanical studies have not clearly identified the desirable coordination of joint actions in the jump shot. Some studies have used linear velocity measurements to study the coordination of shooting (Elliott & White, 1989; Tsarouchas et al., 1990; Walters et al. 1990), while others have used angular velocity measures (Satern, 1990; Satern & Keller-McNulty, 1992). Researchers have observed several types of coordination in shooting a basketball. Walters et al. (1990) reported that four subjects used push actions or the simultaneous action of segments in a jump shot. Other studies have reported several combinations of timing joint actions (Satern, 1990), and more sequential timing of joints as the distance of the jump shot increases (Elliott & White, 1989). Tsarouchas et al. (1990) studied highly skilled free throw shooters from an international tournament. They compared the shooting styles of two players and found that a "high elbow" technique of one player, compared to the typical lower elbow starting position of the other player, diminished the effect of angular motion at the shoulder, thus simplifying the shooting technique.

Some research on the arm action in shooting has demonstrated that

skilled shooters tend to begin the shot with more elbow flexion than less skilled shooters (Satern, 1988; Yates & Holt, 1982). Elliott and White (1989) found a nonsignificant difference (5°) in elbow angle in preparation for the two-or three-point jump shot, but did find that range of motion of the shoulder and wrist increased as the distance of the shot increased, while lower extremity range of motion remained the same. Satern et al. (1989) also observed greater shoulder flexion in adolescents shooting at higher baskets. Research on the free throw demonstrates that the vertical velocity of ball is strongly related to the vertical velocity created by the legs, and the forearm contributes significantly to both horizontal and vertical velocity of the ball (Tsarouchas et al., 1990).

Since there are several effective coordination strategies in skilled shooters, the "smoothness" and integration of the upper and lower extremities may be related to the emphasis coaches place on shooting "rhythm." The upper and lower extremity cooperate to accelerate the ball to a final trajectory appropriate for the shot. Figure 2 illustrates the path of the ball near release in a successful free throw attempt by a male collegiate player with a career free throw shooting percentage of 78 percent. This figure illustrates the nearly linear path of the ball prior to release (Tsarouchas et al., 1990; Walters et al., 1990), which closely approximates the successful trajectory for the shot. Note the smooth change of direction and the linear motion of the ball prior to release. This shot was released at 51 degrees (shaded area).

The research on jump shots suggests that there are several coordination strategies in successful jump shooting. Elliott and White (1989) observed skilled shooters using a preparatory jump, and after a short pause, the forearm and wrist created the major propulsive force for jump shots 15 feet from the basket. They also observed the same subjects using more sequential extension of upper and lower extremities for a 20-foot shot. Walters et al. (1990) also studied skilled shooters from mid-to long-range and found consistent simultaneous (push) type of coordination, where the segments begin extension simultaneously. The best long-range shooters systematically scaled the horizontal and vertical velocities of the ball, whole body center of gravity, and segments to shoot from greater distances. More study is needed to determine what type of coordination may be best for consistent and accurate shooting. Instructors should attempt to correct any gross discontinuities or releases early in the jumping action of the jump shot. The smoothness, or *cooperation of the upper and lower extremities*, may be the hallmark of jump shooting skill. If there is one or more underlying optimal coordinations in shooting a basketball in the dynamic environment of a game, biomechanical research has not identified them.

Coaches should evaluate and promote vigorous wrist action at release.

Ball Rotation

The spin or *ball rotation* applied at release is critical to creating a "soft" shot that may rebound into the basket, if the ball does not pass cleanly through. The backspin applied to the ball in a jump shot serves to decrease the horizontal velocity of the ball if it strikes the rim, or causes the ball to deflect downward if it strikes the backboard (Alexander, 1990). Yates and Holt (1982) found that the amount of ball spin was one of the factors in jump shooting that was statistically related to success. The effect of ball spin on the flight of the ball has been estimated to be less than 5 percent in typical shots, and would tend to increase the shooter's margin for error (Brancazio, 1981). Two techniques related to generating backspin on the ball in a jumpshot are wrist action and the angle of the forearm to the vertical.

When the ball is held with the correct grip and the forearm is aligned vertically, the shooter can apply pure backspin to the ball. The tendency for shooters to let their elbow drift laterally during the shot takes the arm out of alignment with the basket and causes the ball to spin with a sideward component that may deflect it off the rim. Research documenting that skilled players have forearms more aligned with the vertical (Penrose & Blanksby, 1976; Szymanski, 1967) and that shooting success is statistically related to ball rotation and the angle of the forearm at release (Yates & Holt, 1982) suggests that coaches emphasize the importance of the *shooting plane* and *ball rotation.* Ingram and Snowden (1989) suggest that the proper alignment of the forearm in the shooting plane is difficult to achieve if shooters adopt the classic square up stance and shoulder position.

The joint actions that create backspin in a jump shot are a combination of wrist flexion and radio-ulnar pronation (Martin, 1981). Hess (1980) advocated emphasizing wrist action more than arm action in shooting because this fosters quick execution of the shot. Studies have shown that skilled shooters release the ball with the wrist slightly hyperextended (Elliott & White, 1989; Satern, 1990; Tsarouchas et al., 1990). The wrist, however, is flexing rapidly (1260-1564° per second) at release (Satern et al. 1989; Walters et al. 1990).

Coaches should evaluate and encourage vigorous wrist action at release. For a free throw or a typical 15-foot jump shot with a 7-foot height of release, the time of flight is about 1 second and the ball should complete approximately 2 to 3 backward rotations on the way to the basket. A small amount of tape

or paint on one seam of the ball can assist the coach and athlete to more accurately evaluate the rotation on the ball. Any abbreviation of the wrist flexion or pronation in the follow-through should be corrected. An incomplete follow-through may mean that wrist flexion is already slowing at release, rather than peaking to create good ball rotation.

Based on the biomechanics research on shooting a basketball, the following six key teaching points for mid-range to long-range jump shots are proposed. When teaching or coaching the jump shot these concepts should be evaluated and translated into appropriate cue words for feedback to performers.

1. *Staggered stance and a vertical jump.* Biomechanical research has not shown that squaring up to the basket occurs with skilled players. The research suggests that the jump shot stance should be staggered with the shooting side foot forward and the feet approximately a foot apart. The feet may be turned slightly outward, and the trunk may be slightly angled away from the basket in the transverse plane to align the shooting side of the body with the basket. Skilled shooters and successful shots have been related to minimizing the horizontal motion of the body and maintaining a nearly "vertical" alignment of the trunk during the jump.
2. *Shooting plane.* Performers should keep the shooting side of the body aligned with the basket and as close to vertical as possible. Skilled performers and successful shots have been shown to be related to having the shooting forearm lined up with the basket and very close to vertical. The common error of a flying elbow decreases the shooter's ability to accurately direct the shot and apply the correct spin to the ball. Encourage players to "line up" in a plane with the basket.
3. *Optimized height of release.* A high point of release provides a greater margin for error and probability of success. Technique points that are related to the height of release are a balanced jump, flexion of the shoulder at release, and timing the release just prior to the peak of the jump. A cue to stress is "extend at the top of the jump."
4. *Angle of release.* Performers should shoot the ball about 52 degrees above the horizontal. Shooting the ball between 49 and 55 degrees ensures a proper angle of entry and minimizes ball speed. Many cues can get performers to shoot "up" or at the "easiest arc."
5. *Cooperation of the upper and lower extremities.* Research has not clearly identified one style of coordination in the jump shot. Skilled shooters use several patterns of timing joint actions. The jump shot should be an integration of a controlled upper and lower extremity action, without overreliance on either.
6. *Ball rotation.* Vigorous wrist flexion and pronation peaking near release creates backspin on the ball. Backspin helps decrease the speed of the ball if it touches the rim, and may result in a more favorable bounce and a "soft" shot. A typical 15-foot jump shot would make approximately 2 to 3 revolutions on the way to the basket. Emphasize that athletes "flip the fingers" under the ball.

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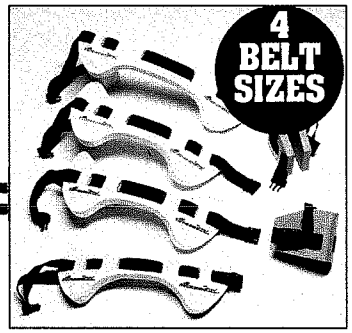

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